Corporate Financial Distress Prediction Using LSTM-GNN

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***Abstract*—Corporate financial distress imposes significant costs on investors, creditors, employees, and the broader economy, making its early and accurate prediction a critical problem in financial risk analysis. Traditional distress prediction approaches typically rely on static, single-period financial ratios and treat firms independently, thereby overlooking two important sources of information: the temporal evolution of a company's financial condition across multiple reporting periods, and the relational structure that exists between companies through shared industry membership and financial similarity. This project proposes a hybrid deep learning framework that jointly models these two dimensions. A Long Short-Term Memory (LSTM) network is used to learn temporal patterns from multiple years of a company's financial history, capturing trends such as deteriorating liquidity or rising leverage that a single-year snapshot cannot reveal. A Graph Neural Network (GNN), built on a company relationship graph constructed from industry and financial-similarity links, is used to learn relational patterns between firms. The LSTM-generated temporal embeddings are then used as node features within the graph, allowing the GNN to combine temporal and relational information into a unified hybrid LSTM-GNN architecture. The proposed system is formulated as a binary classification problem that predicts whether a company is likely to experience financial distress in a future period, and its performance will be benchmarked against classical machine learning baselines – Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting – as well as standalone LSTM and GNN models. The objective of this comparison is to determine whether temporal information, relational information, or their combination meaningfully improves financial distress prediction over traditional approaches. No dataset statistics or experimental results are reported at this stage, as the final dataset and implementation are yet to be finalized.**

***Index Terms*—Corporate Financial Distress, Financial Risk Prediction, Machine Learning, Deep Learning, Long Short-Term Memory, Graph Neural Networks, Financial Ratios, Temporal Data, Graph Learning, Corporate Risk Analysis, Bankruptcy Prediction, Hybrid Neural Architectures**

I. Introduction

Corporate financial distress refers to a state in which a company faces significant difficulty in meeting its financial obligations, often preceding bankruptcy, default, or severe restructuring. The ability to predict financial distress before it materializes is of substantial importance to investors seeking to protect capital, creditors assessing lending risk, regulators monitoring systemic stability, and company management seeking corrective action. Early and accurate identification of at-risk firms can enable timely intervention and reduce the economic damage associated with corporate failure.

Traditional approaches to financial distress prediction, dating back to discriminant-analysis-based models such as the Altman Z-score, primarily rely on financial ratios computed from a single reporting period. While such ratio-based approaches remain useful and interpretable, they suffer from an important limitation: they treat each observation as an isolated snapshot and largely ignore how a company's financial condition changes over time. In practice, financial deterioration is rarely instantaneous – it tends to unfold gradually, through worsening liquidity, rising leverage, or declining profitability across several consecutive years. A model that only observes the most recent period may therefore miss early warning signals that are only visible in the trajectory of a company's financials.

A second limitation of traditional approaches is that they generally analyze companies independently, without accounting for the relationships that exist between firms. Companies operating within the same industry are often exposed to similar macroeconomic and sector-specific shocks, and firms with similar financial profiles may exhibit correlated risk behaviour. Ignoring this relational structure discards potentially valuable information that could improve prediction accuracy, particularly during periods of sector-wide stress.

Motivated by these two limitations, this project proposes a hybrid framework that combines temporal and relational modelling. A Long Short-Term Memory (LSTM) network is used to model the sequential evolution of a company's financial ratios and indicators over multiple years, learning patterns associated with improving or deteriorating financial health. In parallel, a Graph Neural Network (GNN) is used to model relationships between companies, where nodes represent firms and edges represent industry membership and/or financial similarity. The temporal embeddings generated by the LSTM are subsequently used as node features within the graph, allowing the GNN to propagate and refine this information across related companies. This results in a hybrid LSTM-GNN architecture that is, to the best of our knowledge, well-suited to capturing both the temporal and relational dimensions of

corporate financial distress simultaneously.

The remainder of this synopsis outlines the specific objectives of the project, reviews related work in financial distress prediction and graph-based deep learning, and describes the proposed methodology in detail.

II. Objectives

The key objectives of this project are as follows:

- To develop an early-warning system capable of predicting corporate financial distress in a future period using historical financial information.
- To extract meaningful financial ratios – covering liquidity, leverage, profitability, and solvency – from raw financial statement data.
- To analyse historical financial trends and derive temporal indicators that capture whether a company's financial condition is improving or deteriorating.
- To implement and evaluate baseline classical machine learning models, including Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting.
- To implement a Long Short-Term Memory (LSTM) network capable of learning temporal patterns from multi-year financial sequences.
- To construct a company relationship graph based on industry membership and/or financial similarity between firms.
- To implement a Graph Neural Network (GNN), specifically a Graph Convolutional Network (GCN), capable of learning relational patterns between companies.
- To develop a hybrid LSTM-GNN model that combines temporal embeddings with graph-based relational learning.
- To compare the performance of all implemented models – classical ML, LSTM, GNN, and hybrid LSTM-GNN – under a consistent evaluation protocol.
- To evaluate all models using suitable classification metrics, including accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC, with particular emphasis on recall given the real-world cost of missing a distressed company.

III. Related Work

Research on corporate financial distress and bankruptcy prediction spans several decades and has evolved from classical statistical methods to modern deep learning approaches.

Traditional financial ratio-based models. One of the earliest and most influential contributions is the Altman Z-score, which used multiple discriminant analysis on financial ratios to predict corporate bankruptcy [1]. This work established financial ratios – particularly those related to liquidity, leverage, and profitability – as core predictors of distress and remains a widely cited benchmark in the field. Building on this line of work, Ohlson proposed a logistic regression-based probabilistic model for bankruptcy prediction, offering an alternative to discriminant analysis and enabling probability-based interpretation of distress risk [2].

Hazard-model and dynamic approaches. Shumway argued that single-period, static models such as the Z-score can produce biased estimates because they fail to account for the fact that a firm's risk changes over time, and proposed a hazard model that incorporates time-varying covariates [3]. This work is directly relevant to the present project's motivation for incorporating temporal financial trends rather than relying solely on a single-period snapshot.

Machine learning approaches for bankruptcy prediction. More recent work has applied ensemble and boosting-based machine learning methods to bankruptcy prediction, demonstrating that tree-based ensembles such as gradient boosting can outperform classical statistical models on structured financial data [4]. This supports the inclusion of Random Forest and Gradient Boosting as strong classical baselines in the present project.

Recurrent neural networks for financial time series. The Long Short-Term Memory architecture, introduced by Hochreiter and Schmidhuber, was specifically designed to overcome the vanishing gradient problem in standard recurrent neural networks and to retain relevant information over long sequences [5]. LSTM and related recurrent architectures have since been widely applied to financial time-series problems, motivating their use in this project to model multi-year financial trajectories of individual companies.

Graph Neural Networks. Kipf and Welling introduced the Graph Convolutional Network (GCN), a scalable approach for semi-supervised learning on graph-structured data that aggregates information from neighbouring nodes [6]. Hamilton et al. extended this line of work with GraphSAGE, an inductive framework for generating node embeddings on large graphs [7]. These architectures form the foundation of the relational modelling component proposed in this project, where companies are represented as nodes connected through industry and financial-similarity relationships. A broader treatment of GNN architectures and their applications is provided in the survey by Wu et al. [8], which discusses the applicability of graph-based learning to domains involving relational and networked data, including finance.

Collectively, this body of work motivates the central hypothesis of the present project: that combining temporal modelling (via LSTM) with relational modelling (via GNN) can capture information that neither static ratio-based models nor single-modality deep learning models can capture on their own.

IV. Methodology

The proposed methodology follows a structured pipeline, illustrated conceptually below, progressing from data preparation through classical and deep learning models to the final hybrid architecture:

Data Collection → Data Cleaning and Preprocessing → Financial Ratio Calculation → Temporal Feature Engineering → Exploratory Data Analysis → Baseline Machine Learning Models → LSTM-Based Temporal Learning → Company Graph Construction → GNN-Based Relational Learning → Hybrid LSTM-GNN Model → Model Evaluation and Comparison → Financial Distress Risk Prediction

A. Data Preparation

The final dataset has not yet been selected; therefore, no dataset name, source, number of companies, number of observations, or number of years is specified at this stage. The dataset is expected to be structured in panel form, containing multiple companies observed over several consecutive years, which is necessary to construct both temporal sequences for the LSTM and relationship graphs for the GNN. Expected fields include `company_id`, `year`, `industry`, and raw financial statement variables such as `revenue`, `total_assets`, `current_assets`, `current_liabilities`, `total_liabilities`, `total_debt`, `shareholders_equity`, `net_income`, `EBIT`, `interest_expense`, `operating_cash_flow`, and `inventory`, along with a binary `financial_distress_label` target variable.

Preprocessing will include data cleaning, handling of missing values, removal of duplicate records, resolution of inconsistent company identifiers, sorting of observations by company and year, safe handling of division-by-zero cases during ratio computation, treatment of extreme values and outliers, feature scaling/normalization, encoding of categorical variables such as industry, and, if required, techniques to address class imbalance between healthy and distressed companies.

B. Financial Feature Engineering

Meaningful financial ratios will be derived from raw financial statement data, including:

- *Liquidity:* Current Ratio = Current Assets / Current Liabilities; Quick Ratio = (Current Assets − Inventory) / Current Liabilities.
- *Leverage:* Debt-to-Equity Ratio = Total Debt / Shareholders' Equity; Debt-to-Assets Ratio = Total Debt / Total Assets.
- *Profitability:* Return on Assets = Net Income / Total Assets; Return on Equity = Net Income / Shareholders' Equity; Net Profit Margin = Net Income / Revenue.
- *Solvency:* Interest Coverage Ratio = EBIT / Interest Expense.
- *Cash Flow:* Operating Cash Flow Ratio = Operating Cash Flow / Current Liabilities.

In addition, temporal features will be computed to capture change over time, including revenue growth, asset growth, debt growth, change in Return on Assets, change in profit margin, and changes in liquidity and leverage ratios across consecutive years. Exploratory data analysis will examine the distribution of healthy versus distressed companies, the extent of class imbalance, the distribution and correlation of financial ratios, industry-wise distress patterns, and trends in financial indicators over time, in order to justify the use of temporal and graph-based modelling.

C. Classical Machine Learning Baseline

Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting will be implemented as baseline models. Logistic Regression serves as an interpretable statistical baseline; Decision Tree provides a non-linear yet interpretable alternative; Random Forest offers an ensemble approach robust to overfitting; and Gradient Boosting provides a strong sequentially-trained ensemble often effective on structured financial data. These baselines establish a performance reference against which the temporal, relational, and hybrid models will be compared.

D. LSTM Implementation

For each company, financial data from the previous n years will be organized into an ordered sequence and passed through an LSTM to predict distress in the subsequent year:

$$\text{Year } t-3 \text{ data} \rightarrow \text{Year } t-2 \text{ data} \rightarrow \text{Year } t-1 \text{ data} \rightarrow \text{LSTM} \rightarrow \text{Prediction for Year } t$$

The LSTM is expected to learn temporal patterns such as increasing debt, declining profitability, decreasing liquidity, and negative cash flow trends. The proposed architecture consists of an input layer, one or more LSTM layers, a dropout layer to mitigate overfitting, a dense layer, and a sigmoid output layer for binary classification. Hyperparameters such as sequence length, number of LSTM units, dropout rate, learning rate, and batch size will be tuned experimentally during implementation.

E. Graph Construction

Companies will be represented as nodes in a graph, with node features derived from financial ratios, growth indicators, and temporal trends (and, in the hybrid model, LSTM-generated embeddings). Edges will be constructed using industry relationships, where companies belonging to the same industry may be connected, and/or financial similarity, where companies with similar financial characteristics are connected using similarity measures or nearest-neighbour approaches. Supply-chain relationships will not be assumed at this stage, as their inclusion depends on the availability of a dataset that supports such information.

F. GNN Implementation

The initial Graph Neural Network implementation will use a Graph Convolutional Network (GCN), which aggregates and propagates information across neighbouring companies in the graph to learn relational patterns associated with financial distress. GraphSAGE may be explored as an extension if time and scope permit.

G. Hybrid LSTM-GNN Architecture

The central contribution of this project is the hybrid LSTM-GNN model, constructed as follows:

1) Historical financial information from multiple years is provided to the LSTM.
2) The LSTM learns temporal financial patterns and generates a temporal embedding for each company.

3) Companies are represented as nodes in a relationship graph.
4) The LSTM-generated embeddings are used as node features.
5) The GNN processes the company graph and learns relational patterns between companies.
6) The resulting representation is passed through dense classification layers.
7) The system outputs the probability of future financial distress.

Historical Financial Data → LSTM → Temporal Company Embedding → Company Relationship Graph → GNN → Graph-Based Representation → Dense Classification Layer → Financial Distress Prediction

H. Model Evaluation

All seven models – Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, LSTM, GNN, and Hybrid LSTM-GNN – will be evaluated and compared using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC where appropriate. Because failing to identify a financially distressed company carries significant real-world consequences, recall and F1-score will be treated as particularly important metrics. Data will be split chronologically – earlier periods for training, a later period for validation, and the most recent period for testing – to avoid information leakage and to reflect a realistic early-warning deployment scenario. No performance values are reported in this synopsis, as no implementation or experimentation has yet been performed; all such results will be obtained and reported following model development.

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